

- **Inside vLLM: Anatomy of a High-Throughput LLM Inference System** (vLLM Blog, Sep 2025) This provides a comprehensive anatomy of a modern inference engine, including the full request lifecycle (from curl/API call through scheduling, prefill/decode phases, KV cache management, to response). It covers components like scheduling, execution, and serving layers—directly aligning with the "lifecycle of an inference call" and gateway-to-server evolution in your course. Link: <https://blog.vllm.ai/2025/09/05/anatomy-of-vllm.html>
- **Life of an Inference Request (vLLM V1)** (Ubicloud Blog, Jun 2025) — Step-by-step trace of a request through vLLM's OpenAI-compatible API server, tokenization, engine core, scheduling, and generation. It's excellent for understanding production request flows in engines like those discussed (e.g., transitioning from gateway to server). Includes code pointers. Link: <https://www.ubicloud.com/blog/life-of-an-inference-request-vllm-v1>
- **The Lifetime of an LLM inference request on a GPU** (Jan 2026) Directly traces the end-to-end lifetime of a single inference request on GPU (prefill → decode iterations, scheduling, memory ops), with visuals and vLLM context> This is very close to the "lifecycle of an inference call" focus. Link: <https://medium.com/itnext/the-lifetime-of-an-llm-inference-request-on-a-gpu-96354871c70c>
- **LLM Inference: Inside a Fast LLM Inference Server** (Dec 2025) Deep dive into real inference servers (vLLM, TGI, TensorRT-LLM), request flow from input to output, optimizations for production throughput/latency, and how servers handle the generation loop. Link: <https://medium.com/towards-artificial-intelligence/llm-inference-inside-a-fast-llm-inference-server-d551f7bfcc97>
- **Hosting LLMs — From Fundamentals to Scaled Production** (Medium, Dec 2025) — Explains the stateful, memory-bound nature of LLM inference requests (contrasting with traditional stateless ML), lifecycle management for long-running generations, and balancing throughput vs. latency—mirrors the slides' production definition of inference. Link: <https://medium.com/@xiaxiami/hosting-llms-from-fundamentals-to-scaled-production-with-hands-on-tutorial-6598d16810e0>
- **AI Inference Explained: The Hidden Process Behind Every Prediction** (Baseten Blog, Jul 2025) Traces the full request lifecycle (API endpoint → routing → processing → response), including queuing, tokenization, and output handling in production setups. Link: <https://www.baseten.co/blog/ai-inference-explained>

Papers and Surveys on Modern Inference Serving

- **LLM Inference Scheduling: A Survey of Techniques, Frameworks, and Trade-offs** (arXiv/TechRxiv, 2025) A comprehensive survey of LLM serving, focusing on request lifecycle, scheduling (control/data planes), challenges like cold starts/HOL blocking, adaptive batching, and metrics (e.g., TTFT, throughput). It compares frameworks and

discusses production constraints, ideal extension to concurrency and observability. Link: <https://www.techrxiv.org/users/994660/articles/1355915>

- **vLLM: An Efficient Inference Engine for Large Language Models** (Berkeley Tech Report/Thesis, 2025) — Details vLLM's design for high-throughput serving, including request management, PagedAttention for KV cache, scheduling, and lifecycle optimizations—state-of-the-art reference for single/multi-GPU inference baselines. Link: <https://www2.eecs.berkeley.edu/Pubs/TechRpts/2025/EECS-2025-192.pdf>

Additional Production-Focused Guides

- **Step-By-Step LLM Serving Guide for Production AI Systems** (Inference.net, Aug 2025) — Covers building a production stack, including API endpoints, streaming, autoscaling, observability, and request handling aligns with the course's cloud deployment (Modal/Lambda) and future classes on constraints. Link: <https://inference.net/content/llm-serving-guide>
- **LLM Inference Optimization in Production: A Technical Deep Dive (Aug 2025)** Covers production inference realities (memory-first failures, KV-cache as bottleneck), dynamic batching/scheduling for requests, prefill/decode distinction, and why standard ML math doesn't apply - echoes my point about common systems mistakes. Link: <https://medium.com/@vamsikd219/llm-inference-optimization-in-production-a-technical-deep-dive-57eacb81550d>